Perturbations: gamma gates rates not just mean inhibition

nb042 · 2 June 2026 · revising · Gamma-gated sparsity

Abstract

nb025′s rate gap admits a cheap reading: PING fires less because the I-loop delivers more inhibition. This entry forecloses that reading and identifies *what* about the I-stream is doing the suppressing — qualitatively (rhythm vs mean), and quantitatively (which temporal precision is required). Scaffolded by ar009 §Leg 1 item 2.

Methods

Pure inference on the trained nb025 PING baseline (seed 42, $\theta_u = \text{off}$). For each batch the I-population spike tensor $\boldsymbol{s}^I_{\text{base}} \in \{0,1\}^{T\times B\times N_I}$ is recorded from a baseline forward pass, then an override tensor replaces it in a second pass via the nb037 hidden-perturbation hook. The E-population experiences only the override I-stream through W^{IE}; the readout consumes the perturbed E spikes. Mean per-cell I rate is matched to the baseline to four decimals across every perturbation.

Four perturbation families:

1. **Baseline** — no override; trained PING dynamics.
2. **Cycle-coherent jitter** — partition the trial into blocks of length $1/f_\gamma$ ($\approx$ 28 ms at the trained operating point from nb041). For each (trial, block), draw a single Gaussian offset $\Delta \sim \mathcal{N}(0, \sigma^2)$ and shift every I-spike in that block by Δ. Within-burst cross-cell synchrony preserved exactly; only the *placement* of each burst is perturbed. Sweep $\sigma \in \{0, 1, 3, 7, 14, 21, 28, 42, 60, 100\}$ ms.
3. **Phase-shuffle** — per-trial permutation π_b of the time axis applied to all I-cells together: $\boldsymbol{s}^I_{\text{shuf}[t,b,n]} = \boldsymbol{s}^I_{\text{base}[\pi_b(t),b,n]}$. Preserves cross-cell co-firing within a timestep; destroys all phase structure.
4. **Rate-matched Poisson** — per-(trial, cell) Bernoulli with $p = \text{count}_{b,n}/T$. Destroys both temporal and cross-cell structure; tests the g_i variance limit.

Results

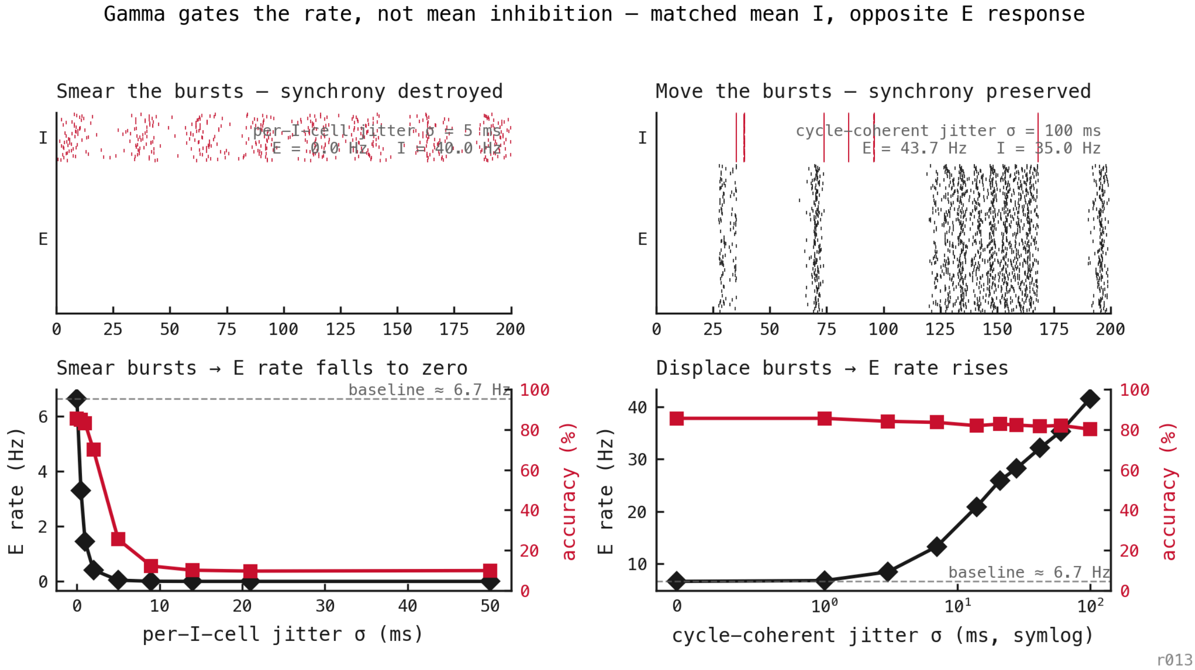

Figure 124: Two inference-time perturbations of the trained-PING I-stream, **both holding mean per-cell I rate fixed** (annotated on each raster). They push the E rate in *opposite* directions, which a mean-inhibition account cannot produce. **Left column — smear the bursts.** Per-I-cell jitter scatters the spikes within each burst, destroying synchrony while leaving the mean untouched; the burst dissolves into a continuous shunt and the E rate *falls to zero*, accuracy collapsing to chance. **Right column — move the bursts.** Cycle-coherent jitter displaces each gamma burst bodily but keeps its within-burst synchrony; the I-stream opens gaps and the E rate *rises* from ≈ 8 Hz toward ≈ 50 Hz as σ grows, accuracy holding near 80%. Same mean inhibition, opposite outcome: what gates the E rate is the *timing* of inhibition — the rhythm — not its average level.

Per-I-cell jitter

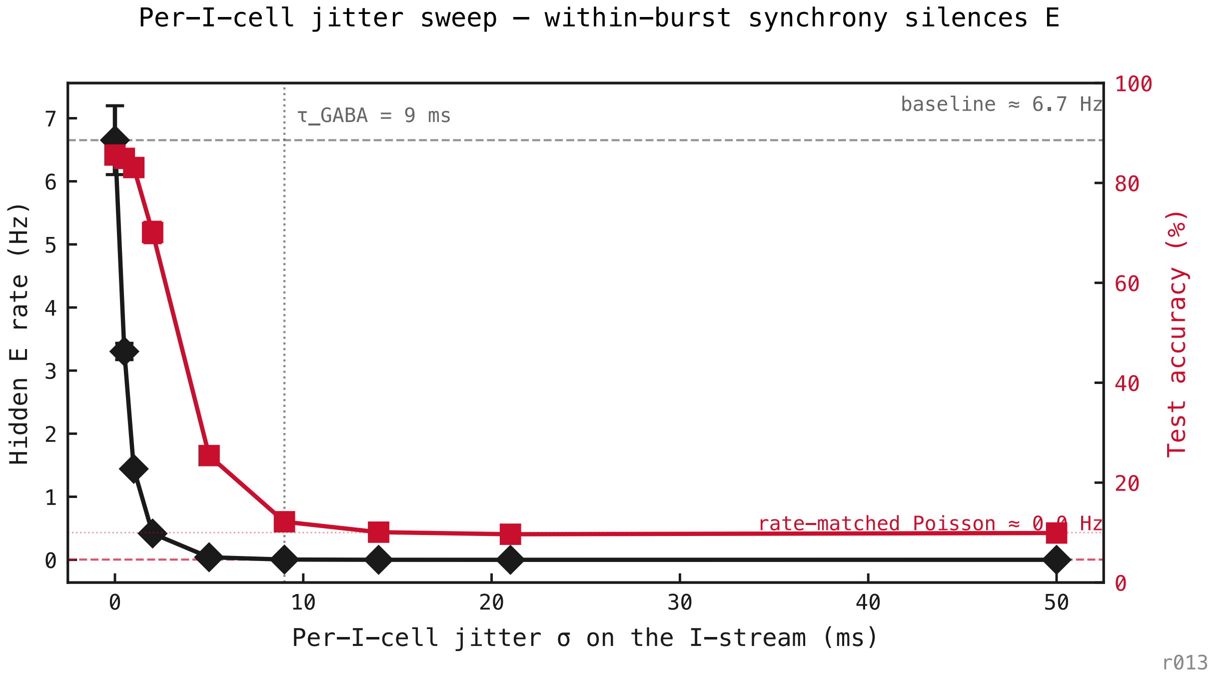

Figure 125: Per-I-cell jitter sweep, three seeds. Each spike receives an independent Gaussian offset; mean per-cell I rate is preserved exactly. **E rate (cyan, left axis) falls monotonically from baseline** — already halved by $\sigma = 0.5$ ms — and is essentially zero by $\sigma \approx 5$ ms, *well below* $\tau_{\mathrm{GABA}} = 9$ ms. **Accuracy (red, right axis) holds at ≈ 85% up to $\sigma = 1$ ms**, then collapses through 71% (σ = 2) and 26% (σ = 5), bottoming at chance (≈ 10%) by $\sigma = 9$ ms.

The $\sigma \to \infty$ asymptote is the rate-matched Poisson regime: E silent, accuracy at chance.

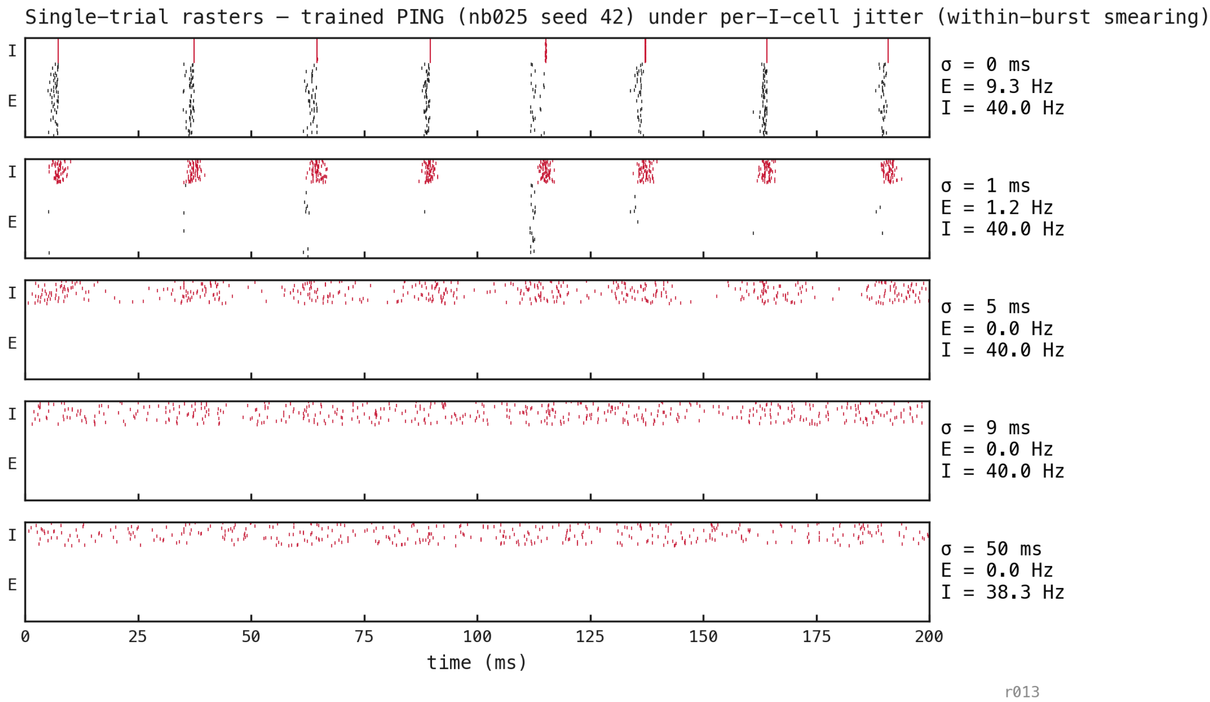

Figure 126: Single trial replayed at five per-cell jitter levels. At $\sigma = 0$ the I-bursts are crisp vertical bands. **At $\sigma = 1$ ms** the bursts visibly smear into a few-ms-wide cluster and E firing already collapses to 1.6 Hz. **At $\sigma \geq 5$ ms** the I-stream looks indistinguishable from a continuous low-variance shunt, and E is silenced. Per-cell jitter doesn't release E; it destroys the bursty structure that gave E recovery troughs in the first place.

Cycle-coherent jitter

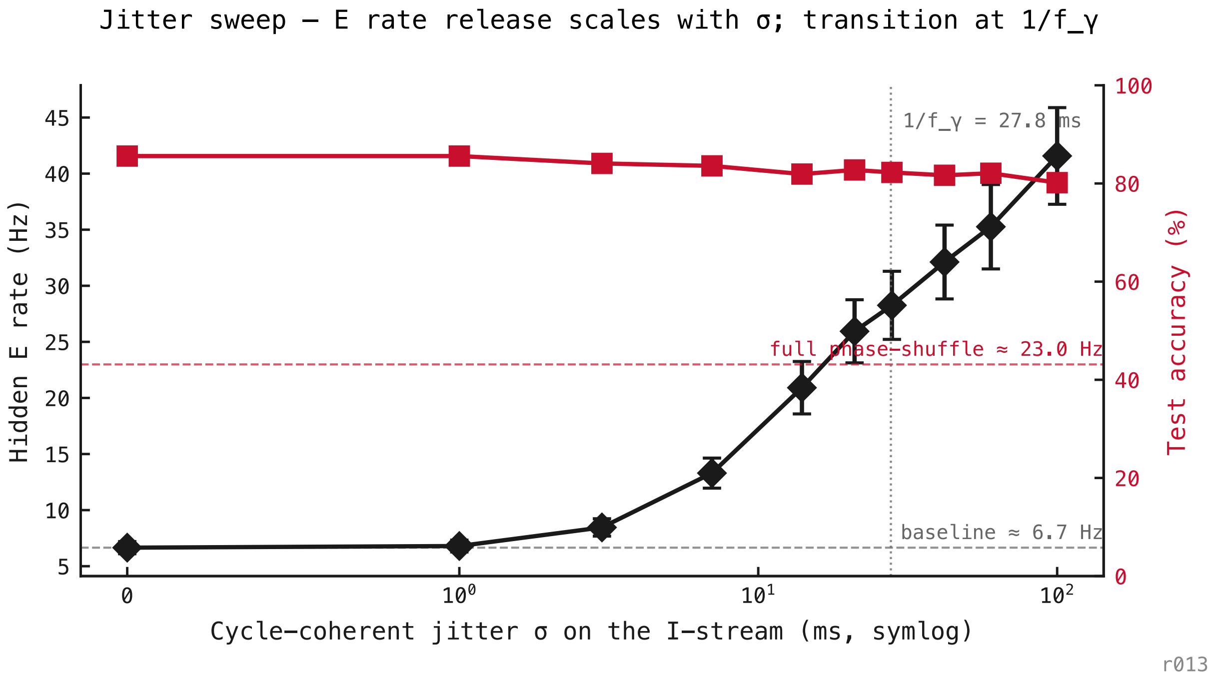

Figure 127: E rate (black) and accuracy (red) vs cycle-coherent jitter σ, seed 42. Horizontal dashed lines mark the baseline rate ($\approx$ 8 Hz) and the full phase-shuffle ceiling ($\approx$ 29 Hz, the $\sigma \to \infty$ asymptote with within-burst structure destroyed). Vertical dotted line at $\sigma = 1/f_\gamma \approx$ 28 ms — the predicted transition timescale.

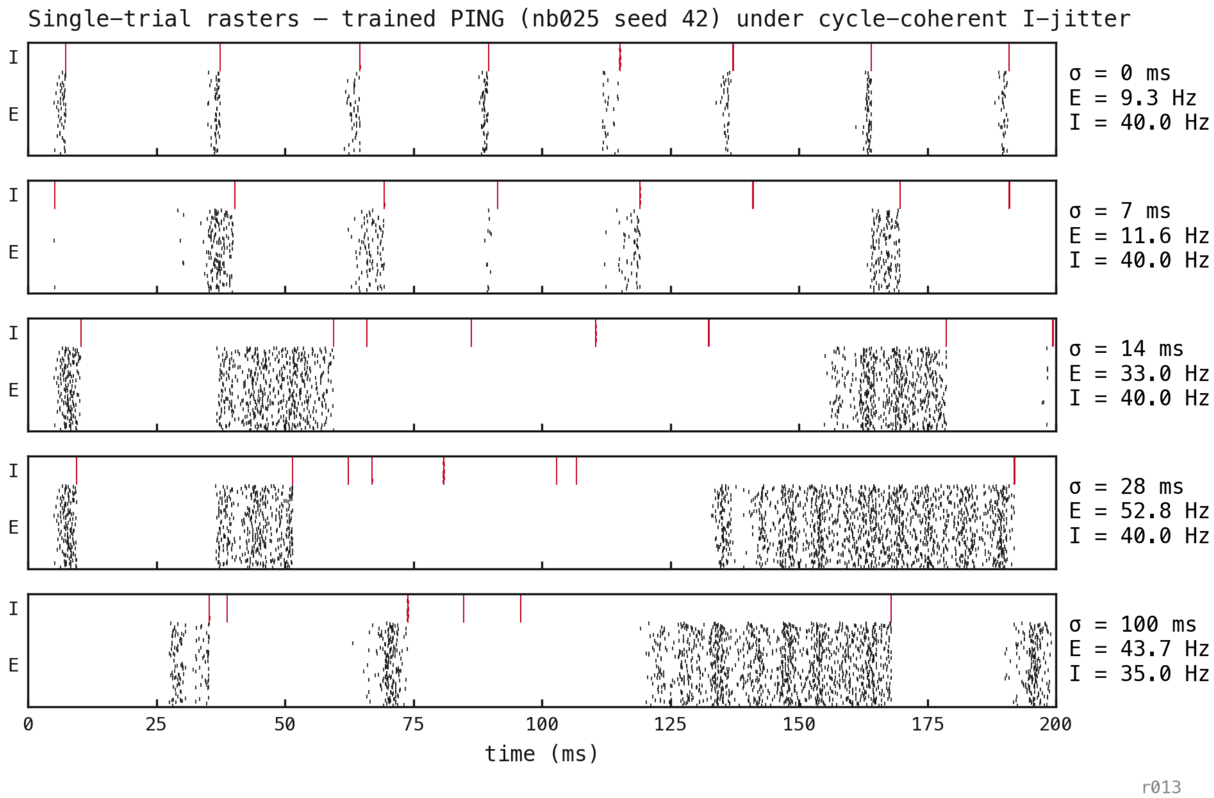

Figure 128: Single trial replayed at five jitter levels (seed 42, MNIST digit 0 sample 0). Per-trial E rate annotated on each panel. **The I-bands stay vertical and crisp at every σ** — within-burst synchrony is preserved exactly. What changes is *where* each burst lands: at larger σ the bursts get displaced bodily from their phase-locked positions, opening longer gaps in the I-stream that E fires through.